THE IDEA

— A ~20 mm mote of printed electronics on biodegradable substrate — seed-dropped over a fuel break or burn scar.
— It sprouts into a temporary sensor skin for fire season, reports through cluster heads, and dissolves.
— Zero e-waste, no retrieval, no grid to build — sensing that ENDS by design.
— Materials/transient-electronics program — fundable R&D, not a product. Every earlier rung must succeed first.

RESEARCH TARGETS

TARGET	GOAL
Mass	<1 g per mote (cluster head <5 g)
Cost	<$1/unit at volume (printed, not assembled)
Life	one fire season (~6 months) then degrade
Degradation	complete in <12 months, non-toxic residue
Comms	duty-cycled LoRa or backscatter — rare TX, local clustering
Power	printed battery + PV/TEG harvest — µW average
Sensing	temp/RH/VOC minimum; optical smoke later

ROADMAP GATES

— Gate 1: university transient-electronics partner (SR&ED eligible).
— Gate 2: lab demo — printed mote senses temp + reports 10 m via cluster head.
— Gate 3: field trial — 100 motes over 1 ha, one season, recoverable data.
— Gate 4: degradable variant — residue confirms <12 month breakdown.
— Gate 5: scale — seed-drop tied to Anchor/Brick infrastructure.

MATERIALS PATHWAY

COMPONENT	CANDIDATE MATERIALS / APPROACH
Substrate	cellulose paper, silk fibroin, transient polymer
Electronics	printed Zn/MnO_2 battery, printed sensors, organic TFT
Energy	printed PV cell, thermoelectric, RF harvest
Comms	backscatter (no TX) or ultra-low-power LoRa wake-up
Deploy	seed pod form factor — germinates sensor on moisture

WHY THIS MATTERS

— Fire season is seasonal — permanent sensors waste 11 months.
— Degradable motes match the threat's rhythm: deploy, vanish.
— Billions-scale is materials R&D, not drone fleet — mesh math supports it.
— Precedent: DARPA/NSF computational-dust programs prove it's real.

HONEST STATUS

— TRL ~2 — concept + literature. No ArcticChain mote exists yet.
— This sheet is a research brief, not a buildable BOM — deliberately.
— Swarm architecture is the same — only the node shrinks.
— Do NOT promise motes to funders until Gate 2 demo exists.

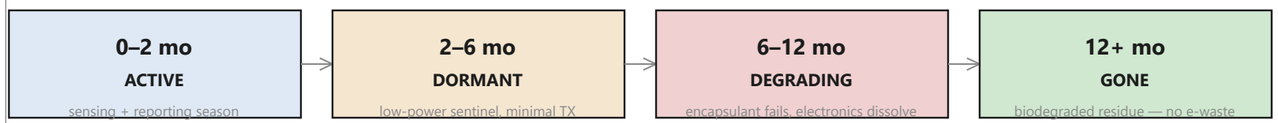
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MATERIALS STACK (TARGET)

LAYER	CANDIDATE MATERIAL	FUNCTION	DEGRADATION
Substrate	cellulose paper / silk fibroin	structure + insulation	weeks–months in moisture
Electronics	printed Zn/MnO_2 + organic TFT	compute + sensing	dissolves with substrate
Energy	printed PV / TEG / RF harvest	µW power	encapsulated, degrades slowly
Comms	backscatter or LoRa wake	rare TX bursts	antenna dissolves last
Encapsulant	shellac / protein coating	weather protection	months — timed release
Seed pod	compressed fiber + nutrient	germination trigger	sprouts sensor on moisture

DEGRADATION TIMELINE



WHY DEGRADABLE MATTERS

— Fire season is seasonal — permanent sensors waste 11 months of their life.
— Degradable motes match the threat's rhythm: deploy, serve, vanish.
— No retrieval mission, no e-waste in wilderness, no environmental liability.
— Seed-drop economics: $0.10–1.00/mote at volume vs $100+ for durable nodes.
— Residue is cellulose/protein — composts in the forest it monitored.

MANUFACTURING PATHWAY

STAGE	PROCESS
Printed electronics	inkjet/aerosol jet — Zn, carbon, dielectric inks
Substrate	cellulose sheet or silk fibroin cast
Energy	printed battery (Zn/MnO_2) or RF harvest coil
Assembly	roll-to-roll print + singulate — no pick-and-place
Seed pod	compressed fiber pellet — moisture germinates

PRECEDENT / FUNDING

— DARPA/NSF 'computational dust' programs — real funded research.
— Transient electronics groups (UIUC, Northwestern, ETH) publish weekly.
— SR&ED eligible — Canadian research tax credit covers 35%+.
— University partners: materials science + EE joint projects.

HONEST GAPS

— No working ArcticChain mote exists — this is a research brief.
— Printed batteries at µW scale are lab-grade, not field-grade.
— LoRa radios don't degrade — antenna/comms is the hard part.
— Cluster heads must be durable — motes feed THEM, not the backbone.
— Gate: lab demo (temp + 10 m report) before ANY field promise.

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UNITS **mm**

NETWORK HIERARCHY

TIER	ROLE	COUNT / DENSITY
Motes	sense temp/RH/VOC — never talk to backbone	thousands–millions per km²
Cluster heads	aggregate mote data, relay to backbone	1 per ~100 motes
Backbone	Anchor/Brick/LTE/sat — the real mesh	existing fleet

Motes don't each shout — they whisper to their cluster head. Density is free, backhaul is precious. Same pattern as satellite constellations.

DUTY CYCLE — ENERGY BUDGET

STATE	CURRENT	DUTY	NOTES
Deep sleep	~1 µA	~98%	printed RTC + wake source
Sense window	~50 µA	~1.5%	10 s every 10 min
Local TX (to head)	~5 mA burst	~0.5%	LoRa or backscatter
Harvest	solar/TEG/RF	continuous	µW average suffices

DATA FLOW

MOTES ×1000
temp/RH/VOC every 10 min → aggregate locally

↓

CLUSTER HEAD
median + anomaly flag → LoRa to Anchor

↓

ANCHOR/BRICK
verify + store-forward → mesh → backhaul

↓

CLOUD/COMMUNITY
dashboard + alerts → EN/FR, OCAP

Only ANOMALIES leave the cluster — raw data stays local. Bandwidth is the constraint; intelligence is the answer.

WAKE-UP METHODS (NO RTC CRYSTAL NEEDED)

— Solar voltage threshold — dawn wakes the field.
— Wake-on-radio — cluster head pings, motes respond.
— Vibration/seismic — fire/vehicle triggers sense window.
— Chemical timer — printed RC slow-release wake.
— Anomaly escalation — mote detects → wakes neighbors to confirm.

OPEN RESEARCH QUESTIONS

— Printed LoRa-class radio — does backscatter suffice at 915 MHz?
— Cluster head durability — can IT be semi-degradable?
— Mote density vs detection probability — modeling needed.
— Biodegradation rate vs fire season length — materials tuning.
— Wake-up reliability — missed windows = missed detections.

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